Zero-Knowledge Zakat: Privacy, Transparency, and Sharia Compliance on the Blockchain

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***Abstract*—Indonesia collects a fraction of its IDR 327 trillion annual zakat potential, not because donors are unwilling, but because existing platforms ask them to sacrifice their dignity to give. Blockchain-based zakat platforms have emerged, but existing systems log donor-recipient transactions on public ledgers, contradicting the Islamic principle of *nafs* (dignity). This paper presents ZKT: a zakat protocol where the blockchain learns only that a valid donation occurred, not who gave, how much, or who received it. Donors select full anonymity or partial disclosure while receiving SBT receipts. On-chain verification provides institutional transparency, but individual transaction data stays private when the donor selects Private mode and the client-side proving pipeline has not been compromised. A Sepolia prototype generates UltraHONK proofs in 246.8 ms (Barretenberg v4.2.1, 16-core Linux) with end-to-end `donateZK()` consuming 721,345 gas. Cryptographic privacy and institutional accountability are compatible, supporting digital zakat that preserves donor privacy; full recipient privacy through encrypted note disbursement remains as future work. The protocol contributes a privacy layer for Islamic social finance that addresses nafs protection under maqasid al-shariah.**

***Index Terms*—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs**

I. Introduction

The Indonesian National Zakat Board (BAZNAS) [1] estimates Indonesia's annual zakat potential at IDR 327 trillion (USD 20 billion), yet actual collection remains lower. Blockchain-based zakat platforms have emerged to address gaps in collection and distribution. However, existing systems log donor-recipient transactions on public ledgers, exposing donor data that contradicts Islamic privacy norms.

Ibn Ashur argued that exposing a person's financial need in public is a harm to their honour, as real as physical harm [6]. A blockchain that records zakat eligibility on Etherscan does exactly that. It publishes poverty. This contradicts maqasid al-shariah directly. Al-Shatibi classified *nafs* (life and dignity) among the five objectives [5], and Thoriquttyas and Rohmawati note that his understanding of Sharīʿah as one that respects human moral agency and social responsibility incorporates dignity" [5]. Islamic sources give concealed charity (*sadaqah al-sirr*) precedence over public giving because it protects dignity.

Blockchain is a decentralized public ledger where transactions are recorded and publicly verifiable. Zakat platforms built on blockchain inherit this transparency by default. Zero-knowledge proofs let you prove something is true without revealing why it is true, which is exactly the property zakat needs. Hameed et al. [7] and Wen et al. [8] have surveyed ZKP constructions and zk-SNARK variants for blockchain deployment. Noir [10] compiles to ACIR, which Barretenberg's UltraHONK prover [9] turns into a proof that is verified on-chain while hiding the witness. Buterin et al. [15] argued that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

This study employs a Design Science Research (DSR) methodology. DSR is appropriate because the contribution is an artifact (the ZKT protocol) whose design must be evaluated against Sharia requirements and performance benchmarks. The protocol implements a Noir circuit encoding nisab and hawl as verifiable predicates, a Pedersen-hash nullifier preventing double-claims, and a four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public).

II. Literature Review

Research on blockchain zakat spans platform implementation, bibliometric analysis, regulatory harmonization, and sustainability assessment. Three implementation papers converge on the same architecture: on-chain records, automated disbursement, public audit trails [2,3,4]. They differ in scope and token design but share a common assumption, that transparency and accountability are the same thing. This paper challenges that assumption. Nazeri et al. [20] examined the potential and challenges. These systems record transactional data on a publicly accessible ledger, making donor-recipient transparency available to anyone running an Ethereum node. Beyond these implementation-focused works, Fauzi et al. [19] confirmed through bibliometric analysis that research interest is increasing across Muslim-majority countries, while Ascarya [20] showed that Islamic social finance instruments (zakat, infaq, sadaqah, waqf) supported economic recovery during COVID-19 and Unal and Aysan [21] proposed blockchain as a harmonization tool for differing Sharia-compliance frameworks. Muharam and Osman [26] explored the sustainability dimension. Evaluated through the maqasid al-shariah framework, all of these systems prioritize *hifz al-mal* (protection of wealth) through audit trails and immutable records, but none addresses *hifz al-nafs* (protection of dignity) through donor-recipient privacy. None of these works integrate privacy-preserving transaction mechanisms.

The three systems differ in scope but converge on the same privacy posture. Khan's platform [2] operates at the collection-and-distribution layer without specifying a token standard or consent model. Nazeri et al. [3] evaluated blockchain features in the Malaysian regulatory context but did not implement a working system. Khairi et al. [4] produced the most complete implementation with ERC-20 tokens, automated disbursement, and an audit trail, yet none of these works incorporate any mechanism for selective disclosure, donor anonymity, or recipient identity protection. Each system treats transparency as a design requirement.

The survey literature covers protocol surveys, tooling and frameworks, and application domains. Hameed et al. [7] covered the protocol space and Wen et al. [8] focused on zk-SNARK variants for blockchain deployment. Williams [21] positioned ZKPs within the blockchain architecture. For tooling, Poseidon, introduced by Grassi et al. [11], reduced constraint counts by up to $8\times$ compared to Pedersen Hash on EVM chains. Several ZKP domain-specific languages exist for writing circuits. Circom, the most adopted, uses an arithmetic-circuit model where constraints are expressed as mathematical equations over finite fields. Noir adopts a Rust-like syntax with type checking and control flow, making circuits readable by auditors who are not cryptographers. Zakat eligibility rules involve nisab thresholds and hawl holding periods that map to conditional logic rather than arithmetic gates, so Noir was selected for this work. This work uses Noir [10], a Rust-like DSL that compiles to ACIR, with Barretenberg's UltraHONK prover generating the proof and a Solidity verifier checking it on Sepolia. The pipeline stays client-side, meaning private inputs never leave the donor's machine. Proof hashes anchor on-chain for anyone to audit. Buterin et al. [15] argued that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

The gap is not technical. ZKP tooling is mature enough. The gap is that no prior work has asked what Islamic law requires of a zakat platform's privacy model. Existing blockchain zakat platforms lack a privacy-preserving mechanism. Nazeri et al. [3] identified traceability as a design objective. General-purpose ZKP frameworks can anonymize transactions but carry no notion of Islamic eligibility. No prior work combines programmable privacy (Noir circuits with verifiable predicates) with Islamic social finance instruments. This paper addresses that gap.

Fig. 1 positions this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology (bottom-right) supports the artifact-driven approach of building, evaluating, and refining the protocol on-chain. ZKT connects these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, instantiated and validated via Sepolia deployment (v8). These gaps motivate the artifact-driven design science methodology described in the following section.

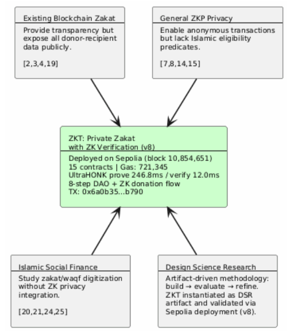

Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

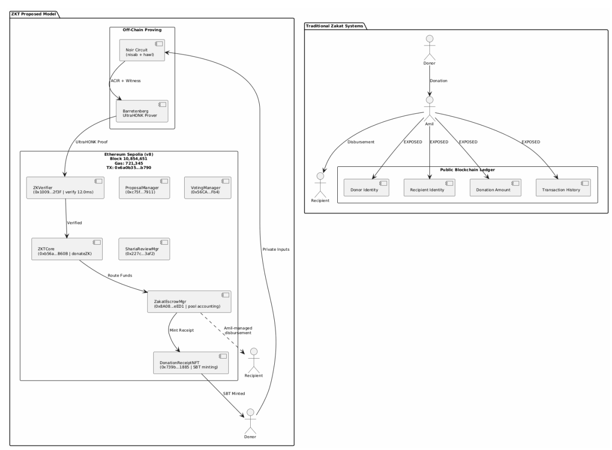

Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. Methodology

DSR was chosen because the research question cannot be answered by proof alone. It requires a working artifact evaluated against Sharia requirements alongside performance benchmarks. A theoretical treatment would leave open whether cryptographic privacy guarantees satisfy Ibn Ashur's conception of dignity. The five DSR phases kept the Sharia evaluation and the performance evaluation on equal footing.

This paper investigates three research questions. RQ1: Can zero-knowledge proofs reconcile on-chain zakat verification with the requirement of donor privacy under Islamic law? RQ2: What protocol architecture achieves this while preventing double-claims through cryptographic nullifiers? RQ3:

How does the resulting protocol satisfy nafs protection under maqasid al-shariah?

The methodology follows five phases (Fig. 3). Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, the Noir circuit architecture encoding nisab and hawl verification, and the four-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v4.2.1, 15 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 15 PWA frontend with wagmi and XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with 247 ms proof generation and 12.0 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis described in Section IV, and Sharia compliance review conducted with input from Tawf Labs domain experts. RQ1 and RQ2 are addressed through Phases 2–4, and RQ3 through Phase 5.

Validity is limited by single-testnet deployment and single-machine benchmarking. The Sharia review involved manual expert assessment, introducing potential researcher bias. The Sepolia deployment (v8) and benchmark data are available at https://github.com/tawf-labs/zkt-hackathon to support independent replication.

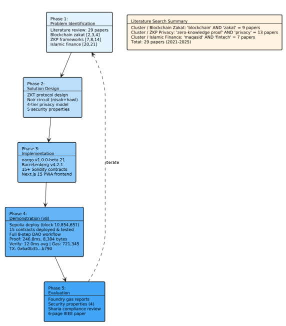

Fig. 3. DSR Methodology with five phases and literature search summary

IV. Results

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model. The actors are donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a malicious claimant, and (3) a public blockchain network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

For the purposes of this protocol, privacy is defined as the inability of an external blockchain observer to link a donation amount to a donor address in Private mode. Accountability is defined as the completeness and public verifiability of the `ZKDonationReceived` event stream and the nullifier registry, such that an amil institution can reconstruct aggregate zakat flow without accessing individual identity records. Institutional trust (*amanah*) is measured by whether the `ProofVerified` event enables independent audit of every claim processed on-chain.

TABLE I
ZKT Protocol Security Properties

Property	Guarantee
Donor anonymity	Identities and amounts never exposed on-chain. Only ZK proofs are publicly verifiable.
Recipient privacy	Donor-side privacy via ZK proofs. Recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data, assuming access to a Sepolia node and independent verification of the Honk verifier contract [5], [6].

The protocol succeeds if four criteria are met. First, no cleartext donor-recipient pairs appear on-chain when the donor selects Private mode. Second, duplicate nullifier submissions are always rejected. Third, proof generation completes client-side, with no private witness data transmitted to an external node. Fourth, amil aggregate compliance reports are independently auditable from emitted events without requiring off-chain data sources.

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 247 ms. The Sepolia settlement layer hosts ZKTCore and ZakatEscrowManager. The Next.js PWA frontend uses wagmi/XellarKit for wallet connectivity.

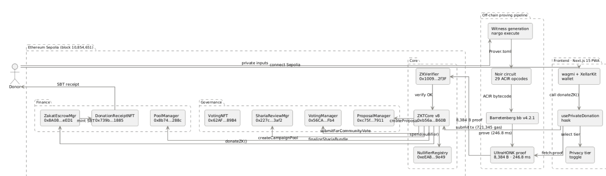

Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

B. System Architecture and Payment Flow

Noir circuits run client-side. Sepolia handles settlement. Donors choose stablecoins or ETH. All donations use `donateZK()`. Proof generation occurs off-chain, while verification runs on-chain. ZK proofs provide privacy. Emitted events and the nullifier registry provide accountability.

The donor uses a Progressive Web Application for wallet connection, privacy tier selection, donation amount entry, proof generation display, and transaction confirmation. A Web Worker performs the proving, relying on browser isolation to keep private inputs on the donor's device [11].

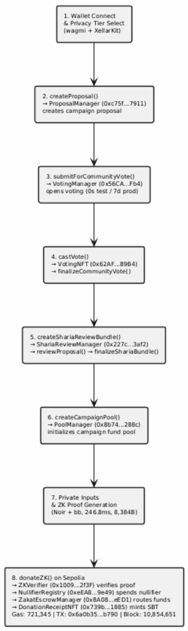

Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

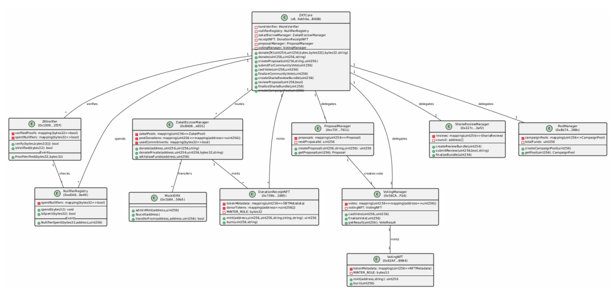

Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v4.2.1 on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end donateZK() gas was measured on the Sepolia testnet at block 10,854,651.

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	246.8 ms
Proof verification (avg)	12.0 ms
Proof size	8,384 bytes

D. Sepolia Deployment and Gas Consumption

15 contracts deployed via single forge script to Sepolia (chain ID 11155111). The ZKTCore (v8, 0xb56a...B60B) routes `donateZK()` through `zakatEscrowManager.donate()` for pool accounting. An end-to-end donation (tx 0x6a0b...b790, block 10,854,651) completed proof verification, nullifier registration, transfer, NFT minting, and event emission in 721,345 gas.

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCore	`donate()`	490,187
ZKTCore	`createCampaignPool()`	365,552
ZKTCore	`castVote()`	132,909
ZKTCore	`donateZK()` (e2e)	721,345
ZKTCore	Deploy	5,314,471
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

The gas measurements establish the protocol's on-chain cost.

E. Security Analysis

Donor anonymity. The only on-chain observable from a private donation is the event emitted by `ZKTCore.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

Private mode zeroes both `donor` and `amount`. No donor or amount data is traceable. Semi-Private attaches the donor's address but hides the amount, a design tradeoff for institutional accountability. Neither mode leaks the full `(donor, amount)` pair. The verifier contract checks a hash and a nullifier flag. These two fields constitute the on-chain footprint of a private donation.

Recipient privacy. The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation $\text{pedersen}(secret, recipient\_address, cycle\_id)$ in `main.nr` means the on-chain nullifier is deterministic but preimage-resistant. An observer sees `0x3f2a...` and cannot recover the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

Circuit soundness. Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient, whose wealth exceeds the nisab threshold as checked by `verify_nisab()` in `main.nr`, cannot produce a witness that satisfies the circuit constraints.

Double-donation prevention. The on-chain `NullifierRegistry` maintains a `mapping(bytes32 => bool)` of spent nullifiers. The `spend()` function enforces single-use through `require(!spentNullifiers[nullifier])`, rejecting a nullifier already recorded. The `ZKTCore.donateZK()` function atomically chains `honkVerifier.verify()` → `nullifierRegistry.spend()` → `zakatEscrowManager.donate()`. A duplicate nullifier causes the transaction to revert. Each eligible recipient can claim zakat at most once per cycle. The nullifier hash is preimage-resistant, so an observer cannot derive the recipient address from it, but the address itself is a public circuit input visible in the transaction calldata. Full recipient privacy, where the amil institution cannot identify claimants, depends on encrypted note disbursement reserved for future work.

Privacy Exposure Analysis. The current prototype exposes several privacy vectors beyond those resolved by the nullifier mechanism. The recipient address is a public circuit input transmitted in the calldata of every `donateZK()` transaction, making it visible to blockchain observers and to the amil institution. In Semi-Private mode, the donor address is emitted in the `ZKDonationReceived` event. Transaction graph analysis on the public Sepolia ledger could correlate donation timing with recipient claim timing. The Web Worker proving model isolates private witness data from the browser main thread but does not protect against IP-level or browser-based side-channel attacks. Donations to the same pool reveal aggregate economic patterns when individual amounts are hidden. Each of these exposures is acknowledged as a limitation of the current prototype and is addressed in the future work roadmap through encrypted note disbursement [15], private decentralized identifiers, and integration with on-chain mixer patterns [16].

The security analysis confirms that the protocol meets its design goals.

V. Discussion

A. Sharia Compliance and Deployment

Maqasid al-shariah has five *daruriyyat*: religion, life, intellect, progeny, and wealth [5]. Ibn Ashur extended *hifz al-nafs* to encompass dignity, arguing that exposing an individual's financial weakness to the public constitutes harm to their honour [6]. A single address recorded as a zakat recipient on Etherscan broadcasts below-nisab status, an outcome that aligns with Ibn Ashur's reasoning.

Azizov et al. [25] propose a maqasid framework for fintech regulation. Latang et al. [24] examine the Islamic legal status of cryptocurrency assets. Pramuka et al. [27] explore blockchain's role in Islamic charitable crowdfunding. This work addresses the donor side, using ZK proofs and nullifier-based anonymity. Recipient-side privacy through encrypted private notes is reserved for future work.

What emerged during evaluation was how the nullifier mechanism maps to Islamic trust concepts. The nullifier prevents fraud while allowing the institution to verify every claim was valid without knowing who claimed it. The cryptography does what an honest intermediary is supposed to do, enacting *amanah*. Private and Semi-Private modes keep donations and linkages off-chain, protecting *hifz al-nafs*. The nullifier registry stops double-claims and the proof's soundness blocks fake eligibility, protecting *hifz al-mal*. The `ProofVerified` event lets amil institutions audit aggregate flows without accessing individual records.

Production deployment requires three prerequisites. Indonesia's OJK Fintech Sandbox must grant regulatory approval, amil institution staff must be trained on wallet management and proof verification workflows, and a rupiah-backed digital currency (such as Bank Indonesia's planned CBDC under Project Garuda) must replace the mock IDRX stablecoin.

B. Limitations and Future Work

The circuit covers nisab and hawl. Eight asnaf categories are possible but not yet implemented, and encrypted note disbursement for recipient privacy is planned as future work. The amil institution disburses through ZakatEscrowManager today with no on-chain proof the money reached the mustahik. DIDs for recipients would close that gap and make disbursements verifiable. A dual governance model, Sharia Council ZK DAO plus Community Donor DAO, is under design. Other directions include recursive proof systems for multi-period eligibility, compliance-friendly ZK protocols for AML/CFT, and batched settlement optimization [7], [17], [23]. These limitations frame the summary of the protocol's findings and its path toward production deployment.

VI. Conclusion

A privacy-preserving zakat protocol was deployed on Ethereum Sepolia using Noir circuits and UltraHONK proofs. The circuit (29 ACIR opcodes) verifies nisab and hawl eligibility client-side, a Pedersen-hash nullifier prevents double-claims, and a four-tier privacy model lets donors choose their disclosure level. This paper makes three contributions: (1) a Noir circuit encoding nisab and hawl as verifiable predicates, (2) a Pedersen-hash nullifier mechanism preventing re-claiming without exposure, and (3) a four-tier donor privacy model (Private, Semi-Private, Verified-Private, Public). Across five benchmark runs, proof generation averaged 246.8 ms (Barretenberg v4.2.1) and an end-to-end Sepolia transaction consumed 721,345 gas.

The deeper finding is that Ibn Ashur's conception of dignity, that financial weakness should not be broadcast to the public, turns out to be a precise specification for what ZK proofs provide. Islamic jurisprudence named this requirement centuries before the cryptography existed to satisfy it. Ibn Ashur's framing of *nafs* protection as encompassing financial dignity [6] aligns with ZK proofs, which verify eligibility while keeping identity, income, and assets private. The `ZKDonationReceived` event gives amil institutions an auditable record of aggregate zakat flow, while the nullifier registry prevents abuse, decoupling what the blockchain records from what it reveals about individuals.

Future work extends the circuit to all eight asnaf categories, adds recipient-side encrypted note disbursement, and pursues OJK Fintech Sandbox integration for production deployment.

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